Multi-Modal Large Language Model with RAG Strategies in Soccer Commentary Generation

Xiang Li*
Southern University of Science and Technology
lixiang960927@gmail.com

Yangfan He*
University of Minnesota-Twin Cities
he000577@umn.edu

Shuaishuai Zu
KNQ.AI
Renmin University of China
zushuaishuai@ruc.edu.cn

Zhengyang Li
KNQ.AI
Imperial College London
zhengyang1835@gmail.com

Tianyu Shi✉
University of Toronto
ty.shi@mail.utoronto.ca

Yiting Xie
KNQ.AI
xieyitingdemon@gmail.com

Kevin Zhang
KNQ.AI
kevinzyz@knqtech.com

Abstract

As a globally celebrated sport, soccer has seen its appeal greatly amplified by engaging and vivid commentary. Recently, Multi-Modal Large Language Models (MLLMs) have attracted attention in generating soccer commentaries due to their remarkable capacities of understanding different modalities of the input videos. Most of these methods have shown that the use of multiple modalities can enhance the commentary quality, which includes video, audio, and structured meta-data. However, delivering precise and rich commentary requires the ability to accurately discern subtle differences in similar backgrounds, events, and players. This presents a significant challenge for existing MLLMs. So we propose SoccerComment, a framework for generating soccer commentary that integrates MLLMs with Retrieval-Augmented Generation (RAG) strategies. This framework enhances inference efficiency and reduces the need for continuous training through a multi-modal clustering memory unit and retrieval-augmented in-context learning mechanisms, ultimately improving the accuracy and diversity of the commentary. Based on similar retrieved scenarios, SoccerComment demonstrates outstanding zero-shot performance, offering a new direction and scalable solution for future research in soccer commentary generation.

1. Introduction

Soccer is one of the world's most popular sports. In recent years, the rise of digital sports broadcasting has led to significant advancements in both the quantity and quality of sports data, offering a solid foundation for automated soccer analysis [6, 14, 17, 21, 27]. Video captioning, the task of describing given videos in concise and accurate language, is a crucial yet challenging task in Computer Vision (CV) and Natural Language Processing (NLP) [23]. Differing from video captioning, soccer commentary needs to contain more fine-grained video descriptions, and the task generally faces the following challenges: 1) understanding video content and their interactions requires linking several objects (e.g., players, goalkeepers, referees, and balls) with corresponding entities in each video frame; 2) synthesizing soccer commentary with coherent event descriptions, motivation inference, and stylized emotional expression suitable for the current event.

Recently, numerous researchers have leveraged MLLMs to deliver concise and engaging soccer commentaries [15, 18, 35, 36]. Video understanding in sports utilizes MLLMs to interpret and analyze complex scenarios beyond the capabilities of existing classification models. Unlike predefined task-based models, MLLMs models can perform complex and open-ended tasks, while CV classification models are more suited for identifying specific and well-defined movements performed by a player. Moreover, compared with Video Large Language Models (VLLMs) only utilizing video frames, MLLMs can attend to information from more modalities, including image, audio, and text, thus reducing the difficulty of generating soccer commentary. Most of these MLLM-based methods demonstrate that the incorporation of multiple modalities can enhance the quality of generated commentaries significantly [33, 35]. How-

*Equal Contribution, ✉Corresponding Author

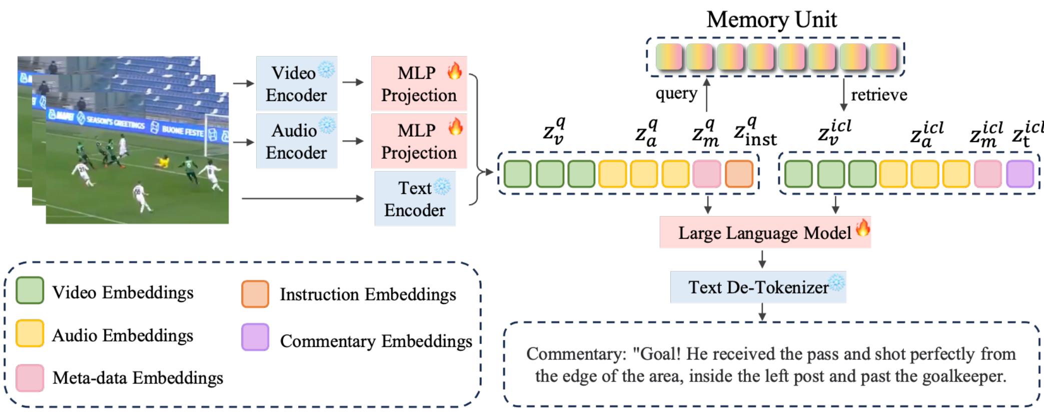

Figure 1. **SoccerComment Overview:** The overall architecture consists of two main components: 1) a Multi-Modal Large Language Model (MLLM) backbone responsible for perceiving multi-modal information from the video and generating commentary text; 2) a memory unit, which is used for clustering and retrieving multi-modal embeddings. These two components are bridged by a retrieval engine to enable the Retrieval-Augmented In-Context Learning (RA-ICL) process.

ever, aligning the multi-modal features to preserve the semantic similarity still remains as a challenge, which limits the efficient information extraction from other modalities.

In this paper, we introduce SoccerComment, a novel retrieval-enhanced MLLM, which takes the multi-modal features of soccer matches as input and outputs the corresponding high-quality commentaries. Specifically, we propose to apply a multi-modal clustering memory unit in the framework, which is an innovative approach in the field of sports video understanding. The Retrieval Augmented Generation (RAG) embedding in the latent space across three modes (video, audio, and text) is constructed by contrast learning for training the multi-modal clustering network, and the case retrieval is carried out in the multi-modal joint cluster. It can effectively recall similar soccer game scenes, along with the wording and style of commentary at the time in a zero-shot setting. Combined with the multi-modal clustering memory unit, the top-k similar soccer video clips are recalled and then transmitted to the MLLM. The Retrieval-Augmented In-Context Learning (RA-ICL) mechanism is introduced to enhance the context information, and then the MLLM can "learn" and "summarize" the paradigm of retrieved video descriptions to generate more accurate commentary.

This paper makes contributions in three main aspects:

- For the task of soccer commentary generation, we propose a novel retrieval-enhanced context learning method based on a Multi-Modal Large Language Model (MLLM), which can enhance the diversity and stylization of model generation capabilities.

- We show that by constructing the multi-modal joint feature space, the retrieval effect of instances can be significantly improved, and the interpretation generation performance of downstream tasks can be enhanced to a large extent.

- We extracted background commentary audio from the SoccerNet-v2 dataset [9] and organized it into text, expanding the multi-modal attributes of the original dataset and providing convenience for future research.

2. Related Work

2.1. Multi-Modal Soccer Understanding

Soccer, as one of the globally beloved sports, has attracted many researchers for automatic soccer analytics, including action spotting [28], player tracking [28], video captioning [29], commentary generation [7], etc. The multi-faceted aspects of soccer analytics are explored through the SoccerNet dataset and challenges [9, 16]. Live commentary generation is the task that provides a rich summary of the game to increase fan engagement for those who do not have time or lack accessibility to watch the game. Recently, there has been a growing interest in the research community in automatically generating commentary text based on corresponding videos [15, 33, 35]. Most of these methods demonstrate that the use of multiple modalities (including but not limited to videos, images, text, audio, and structured metadata such as commentary text, event information, and team statistics) can enhance game understanding. The GOAL [33] integrates video clips with abundant background knowledge, including structured and unstructured knowledge from professional sports platforms and Wikidata, increasing the performance of the video captioning task. MatchTime [35], a high-quality soccer commentary dataset is proposed by utilizing a multi-modal tempo-

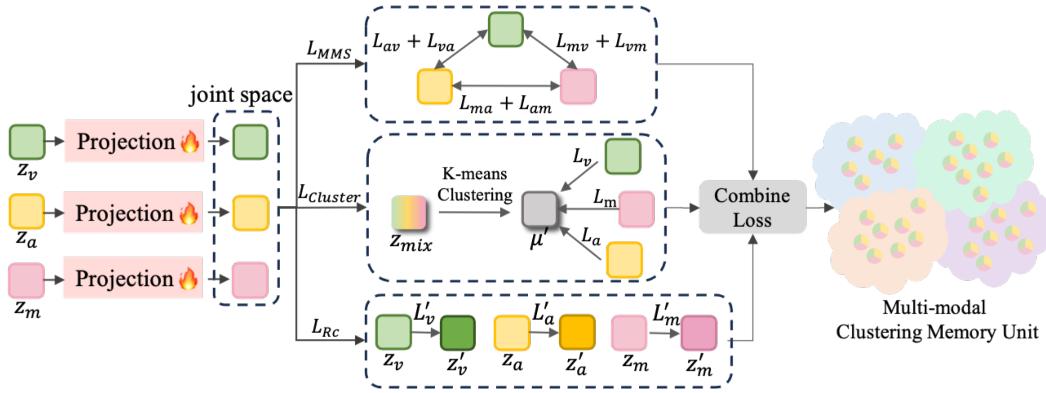

Figure 2. The Multi-modal Clustering Memory Unit is built in four steps: 1) features are mapped to a retrieval space. 2) MMS loss is used to increase similarity within the same instance and decrease it between different instances. 3) K-means clustering groups mixed-modal instances, aligning single-modal representations with their cluster centers. 4) Reconstruction loss ensures regularization. This results in a Memory Unit that combines visual, audio, and text data from query instances for similarity matching and retrieval.

ral alignment pipeline. Based on the curated dataset, the authors developed an automatic soccer game commentary LLM-based model named MatchVoice, which leverages a pre-trained visual encoder and language model to generate accurate and professional commentary text. However, given that the features from different modalities are often not comparable, existing models generally utilize visual information corresponding to each video instance for generating commentary [32]. In this paper, we incorporate audio information and meta-data into our generation process, which can reduce the difficulties of generating commentary just from image frames.

2.2. Multi-Modal Retrieval Augmentation Generation

Retrieval Augmented Generation (RAG) systems [13] mark a significant paradigm shift in Large Language Models (LLMs), combining document retrieval with generative modeling to produce contextually relevant answers. The role of vector databases [19] within RAG systems is pivotal, especially for the efficient management and retrieval of high-dimensional vector representations of text, highlighting their critical contribution to the precision and relevance of the responses generated. RAG has found application in different domains including enterprise contexts where LLMs are integrated with chatbots, enabling them to automatically derive more accurate answers from knowledge bases. In pursuit of more scalable and modular knowledge integration, recent concurrent studies have delved into Multi-Modal Retrieval Augmentation Generation (MRAG) [3, 4, 12, 20], empowering a foundational multi-modal model to access pertinent text and images sourced by a retriever from external memory banks. Re-Imagen [4] performs diffusion-based caption-to-image generation using retrieved images; MuRAG [3] performs natural language question answering using retrieved images. Another important paradigm of LLM is In-Context Learning (ICL) [10]. This paradigm involves providing a test query alongside a few demonstration examples as contextual information. The LLM then generates an output for the test instance based on analogies drawn from context, without any updates to its parameters. Notably, RAG is a systematic approach to curate ICL examples.

As for MRAG, the core difficulty is the alignment of the feature representations from different modalities [20], where these feature representations are required to be projected into a common multi-modal embedding space. In this paper, we learn across three modalities (video, audio, text) in a joint space. Then, we retrieve relevant information from other video instances for leading the LLM to generate more accurate soccer commentaries.

3. Methodology

SoccerComment, a MLLM with RAG strategies, utilizes three types of aligned information for end-to-end soccer commentary task from video inputs. As shown in Fig. 1, SoccerComment comprises two main components: a multi-modal unified perception and generation network based on a MLLM backbone, and a memory unit constructed on a database obtained through hybrid vector clustering. These two parts interact through a retrieval engine to acquire rich multi-modal contextual knowledge, enabling robust in-context learning in the process of commentary generation. Following the successful MLLM paradigm of Video-LLaVA, we first achieve the alignment of visual, audio, and text through instruction tuning. Utilizing a pre-trained video encoder, audio encoder, and sensing modules followed by an LLM, we construct a fully differentiable MLLM by aligning these embeddings through trainable MLP projectors.

3.1. Video and Audio Encoder

The video and audio encoder is derived from LanguageBind [41], based on ViT-L/14 [11]. We segment the entire soccer match into multiple video frame sequences $V_i = \{v_i^1, v_i^2, \ldots, v_i^{vk}\}$ and audio sequences $A_i = \{a_i^1, a_i^2, \ldots, a_i^{ak}\}$ according to the commentary timestamps, where vk and ak represent the number of video frames and spectrum frames contained in this video instance (slice) respectively. These sequences are then transformed to output video embedding $\mathbf{z_v}$ and audio embedding $\mathbf{z_a}$.

3.2. Large Language Model Backbone

We use Vicuna-7B v1.5 [40], which is instruction-tuned based on LLaMA2 [37], as the LLM backbone. The LLM takes the aligned $\mathbf{z_v}$, $\mathbf{z_a}$, language token embeddings $\mathbf{z_m}$ (meta-data from sensing module), and $\mathbf{z_t}$ (instruction) as input. As a decoder-only LLM, each output token x_n is sampled auto-regressively based on the previous output x_{n-1} and the multi-modal contextual prefix $\mathbf{z_{1:n}} = [\mathbf{z_v}, \mathbf{z_a}, \mathbf{z_m}, \mathbf{z_t}]$, and finally decoded into language space through a text detokenizer.

3.3. Multi-Modal Clustering Memory Unit

Another crucial part is constructing a memory unit based on a joint representation space. Each video instance is associated with its corresponding single-model information (including visual representation, audio, meta-data, and commentary text), and proximate to instances with similar semantics, while distant to those with significant content differences. Then, as demonstrated in Fig. 2, we follow the work by Ilharco, Zhang, and Baldridge [22], introducing a contrastive loss L_{MMS} to compare the single-modal representations of each video instance, a clustering loss L_{Cls} to make single-modal information which has similar semantics remain close, and a reconstruction loss L_{Rc} to facilitate a more stable cluster training. These three losses are used to guide the optimization direction of the model.

3.3.1 Contrastive Loss for Joint Spaces

We employ the Masked Margin Softmax (MMS) Loss [22] as the loss function, which represents the similarity between modalities based on the dot product of embeddings. Unlike traditional triplet loss where positive and negative samples need to be explicitly specified and only one negative sample from one batch is used, the MMS loss leverages all negative samples within the batch, thereby improving sample utilization efficiency. To learn the joint space of three modalities, we calculate the contrastive loss for all modality pairs and add them to get the total contrastive loss L_{MMS},

$$L_{MMS} = (L_{vm} + L_{ma} + L_{av}) + (L_{mv} + L_{am} + L_{va}) \quad (1)$$

where L_{vm}, L_{ma}, L_{av} represent the loss associated with pairwise modalities $(\mathbf{z_v}, \mathbf{z_m})$, $(\mathbf{z_m}, \mathbf{z_a})$ and $(\mathbf{z_a}, \mathbf{z_v})$ respectively, and L_{mv}, L_{am}, L_{va} represent the pairwise losses in opposite directions. Take L_{vm} as an example, one component of the MMS loss is given by:

$$L_{vm} = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{e^{\mathbf{Z}_{v_i,m_i} - \delta}}{e^{\mathbf{Z}_{v_i,m_j} - \delta} + \sum_{j=1}^{B} \mathbf{M}_{v_i,m_j} e^{\mathbf{Z}_{v_i,m_j}}} \quad (2)$$

where B is the mini-batch, $\mathbf{Z}$ is the similarity matrix of all samples within the batch ($\mathbf{Z}_{v_i,m_j}$ denotes the pairwise similarity between the v_i embedding and m_j embedding in the joint latent space). The mask matrix $\mathbf{M}$ is used to identify positive pairs and filter negative pairs, where $\mathbf{M}_{v_i,m_j} = 0$ refers to the positive pair and $\mathbf{M}_{v_i,m_j} = 1$ refers to the negative pair. L_{vm} represents the loss for a given video sample paired with negative text samples enumerated. On the other hand, L_{mv} is the loss for a given text sample paired with various negative video samples. δ is a margin hyperparameter that is empirically selected, to control the difficulty of contrast learning.

The aim of L_{MMS} is to maximize the similarity between modal pairs belonging to the same instance while widening the distance between modal pairs belonging to different instances. This formula for pairwise contrast loss ensures that features between different modes are comparable.

3.3.2 Online Clustering and Semantic Learning

We choose to use K-means clustering to perform unsupervised clustering for multi-modal features, ensuring that semantically similar instances are close to each other in the joint representation space. The process consists of two main parts: mixed-modal clustering and single-modal semantic learning.

The mixed semantic modality combines three modalities [2]:

$$\mathbf{z_{mix}} = (\mathbf{z_v} + \mathbf{z_m} + \mathbf{z_a})/3 \quad (3)$$

in our case, we take the mean over these embedding to represent a multi-modal instance, which is used as the input to the k-means algorithm, producing k dispersed clusters characterized by the centroid matrix $\mathbf{C} = \{\mu_\mathbf{1}, \ldots, \mu_\mathbf{k}\}$. The cluster assignments for $\mathbf{z_{mix}}$ are defined by solving the following problem:

$$\min_{\mathbf{C} \in \mathbb{R}^{d \times k}} \frac{1}{N} \sum_{n=1}^{N} \min \|\mathbf{z_{mix}} - \mathbf{C}\,\mathbf{h_n}\|_2^2 \quad (4)$$

where $\mathbf{h_n} \in \{0,1\}^k$ is the column vector to represent the cluster to which sample n belongs. To prevent the algorithm from falling into local optima, we use the videos that belong to the contextual label of the SoccerNet dataset as 13 initial cluster centers (12 categories from SoccerNet and one alternate category) for initialization. An online incremental update method is adopted to ensure that the clustering process acquires sufficiently rich semantic information.

After obtaining the clustering results based on mixed modalities, we construct a loss function that treats the cluster centers as targets, forcing the single-modal data to approach their respective mixed-modal cluster centers while distancing themselves from other cluster centers. Then, we sum the losses of the three single modalities to obtain L_{Cls}:

$$L_{Cls} = L_v + L_m + L_a \tag{5}$$

And components of this loss, for example, the video modality, L_v is given by:

$$L_v = -\frac{1}{B}\sum_{i=1}^{B} \log \frac{e^{\mathbf{z_{ti}}\cdot\mu' - \delta}}{\sum_{k=1}^{K} e^{\mathbf{z_{ti}}\cdot\mu_{\mathbf{i}}}} \tag{6}$$

where μ' is the nearest cluster center for $\mathbf{z_{ti}}$. The similarity of the sample to the correct clustering center is maximized and the similarity to other clustering centers is minimized. During inference, the projected single-modal data will find the nearest cluster center as well as other semantically similar instances.

3.3.3 Multi-modal Reconstruction

Feature reconstruction is an auxiliary task in contrast learning, which helps to standardize training and improve model generalization ability. In order to stabilize cluster training and help capture features suppressed by contrast learning and clustering, we add reconstruction losses to the space of the three single modalities:

$$L_{Rc} = L'_v + L'_m + L'_a \tag{7}$$

And the component $L_{v'}$ is given by:

$$L_v = -\frac{1}{B}\sum_{i=1}^{B} \|\mathbf{z'_{vi}} - \mathbf{z_{vi}}\|^2 \tag{8}$$

where $\mathbf{z}'$ is the reconstructed features by feeding $\mathbf{z}$ into two linear layers as encoder and decoder.

3.4. Retrieval-Augmented In-Context Learning

RA-ICL bridges the MLLM and the memory unit. Encoders of MLLM output query features $\mathbf{z^q}$ and the nearest cluster center μ' and its semantic category is determined by K-means clustering. The cosine similarity between $\mathbf{z^q}$ and each embedding in the database is computed as $S(\mathbf{z^q}, \mathbf{z_i^{icl}}) = \frac{\mathbf{z^q}\cdot\mathbf{z_i^{icl}}}{\|\mathbf{z^q}\|\|\mathbf{z_i^{icl}}\|}$. We consistently select the most 2 relevant retrieved instances based on this similarity score. In order to perform the RA-ICL, we prefix these retrieved instances before the query, facilitating an implicit gradient descent through the meta-optimizer of LLM, as proved by Dai *et al.* [8]. RA-ICL enhances the prediction and generalization ability of the model by using the relevant examples retrieved without additional training.

4. Experiments

4.1. Settings and Datasets

4.1.1 Data Input

From the open source SoccerNet-v2 dataset [9], we obtained 471 matches with title tags. We used all the videos in 25fps and 720p resolutions and cropped a 30-second video of each event (15 seconds before and after the event) based on the timestamp in the annotation. In the end, we obtained 32,253 captioned match segments. In addition to using the original SoccerNet-V2, we also utilized the corrected time annotations from MatchTime [35] to guide the video clipping. If the video segment has been corrected by MatchTime, the corrected time is used; otherwise, the original time is applied. In Table 1, we compared the results of training and evaluation using different datasets, and in subsequent ablation experiments, we use the MatchTime corrected dataset.

4.1.2 Implementation Details

The video encoder and audio encoder we used are both the Lora-tuned versions of LanguageBind-Video and LanguageBind-Audio. The fine-tuning MLLM uses a learning rate of 2e-5 and the warm-up ratio is 0.03 for 5 epochs. All experiments were conducted with 8 Nvidia A800 GPUs.

We used the VideoFileClip method from the MoviePy library to extract the audio modality. For the meta-data, we removed redundant information from the SoccerNet-v2 [9] caption annotation, keeping only key details such as the match venue, match time, player names, and player numbers. We also added the transcription of the audio into the meta-data. The specific process is as follows: we use Whisper-large-v2 [34] to extract and translate the audio into English, then use the ChatGPT-4 [30] model to organize and refine the translated English. A specific example of the meta-data can be found in Fig. 3.

The whole training process includes three stages: MLLM training, clustering training, and RAG finetuning. In the MLLM training stage, we froze the encoders and only trained the MLP projection and the LLM. This projected the pre-trained video embeddings and audio embeddings into language tokens understandable by the LLM, aligning the visual, audio, and text features. Next, we used the embeddings after the encoder and MLP projection for clustering space training. Finally, in RAG finetuning stage, we retrieved the semantically closest instances from the cluster-

Table 1. Comparison of state-of-the-art methods and SoccerComment.

Methods	BLEU-1 ↑	METEOR ↑	Rouge ↑	CIDEr ↑	Sentiment Score ↑
GOAL [33]	15.60	6.40	11.50	6.60	75.2
MatchVoice [35]	31.42	<u>26.12</u>	29.66	**38.42**	80.3
SoccerNet-Caption [29]	29.74	**26.40**	26.19	23.74	77.8
Video-llama [39]	19.13	15.24	12.16	3.44	<u>89.0</u>
InternVL [5]	21.21	14.86	16.57	4.47	88.2
Video-chatgpt [26]	14.02	21.95	12.93	2.77	**94.8**
Video-llava [24]	18.83	19.18	14.28	3.27	87.2
SoccerComment†	<u>31.80</u>	18.21	<u>30.41</u>	35.43	83.4
SoccerComment‡	**33.42**	18.98	**31.78**	<u>36.58</u>	85.5

†and ‡represent training on original SoccerNet-Caption and Matchtime dataset respectly.

we denote the best performance in **bold** and the second-best performance in <u>underline</u>.

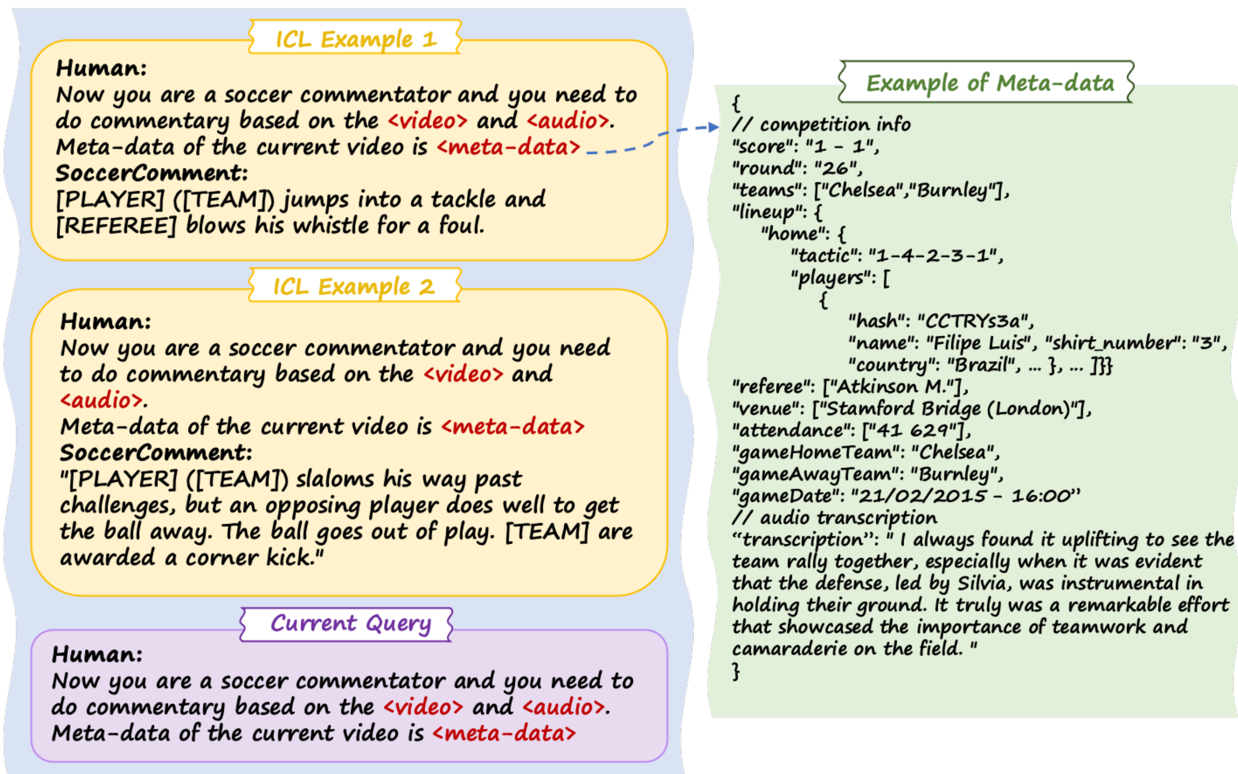

Figure 3. Example of query and ICL. The query example includes the current query, which contains the question and answer instruction, video embeddings (**<video>**, audio embeddings (**<audio>**) and text embedding **<meta-data>**. The retrieved ICL instances are used as a prefix for the query and together fed to LLM.

ing space using the RAG strategy, and finetuned the MLLM jointly with the query instance.

4.1.3 Benchmark Setting and Evaluation Metrics

To ensure fairness, we compared models that are specifically designed for soccer commentary generation and general video understanding models that have not been finetuned, and used the test set officially separated by SoccerNet-V2 [9] to evaluate the results. Except for common metrics (BLEU [31], METEOR [1], Rouge [25] and CIDEr [38]), we also adapted Sentiment Score (which evaluates the sentiment polarity (positive, negative, neutral) and intensity of the text using pre-trained sentiment analysis

Figure 4. Examples of Soccer Commentary Generation from SoccerComment.

models, such as BERT-based sentiment analysis models) for evaluation.

4.2. Results Analysis

We compared several soccer commentary generation models, general video understanding models, and multimodal large language models (MLLMs). The results demonstrate that SoccerComment outperforms most of the models across various metrics (see Table 1). To provide a more intuitive visualization of the efficacy of our framework, Figure 4 displays a comparison of the raw video frames, the output of the MLLM model (predictions), and the ground truth annotations.

Specifically, SoccerComment achieves state-of-the-art (SOTA) performance in BLEU and Rouge, with improvements of 5.09% and 4.51%, respectively, compared to the second-best model (which is also a version of SoccerComment trained on SoccerNet). This demonstrates that the generated text has higher semantic consistency with the annotations, and the structure and content of the commentary are more complete. The enhancement through RAG leads to more accurate commentary. SoccerComment ranks second in CIDEr, with MatchVoice performing the best, indicating that the generated results by SoccerComment are closer to real-world expressions and exhibit higher diversity. In terms of METEOR, SoccerComment performs relatively lower compared to other models. This is due to some differences in length and grammar between the generated commentary and the annotations. There is room for further improvement in language fluency and grammatical structure, which is an area we will focus on optimizing in the future. Regarding sentiment scores, SoccerComment scores lower compared to general models, but higher than MatchVoice and SoccerNet-Caption. We believe this is because the fine-tuned LLM model used in SoccerComment, after incorporating meta-data and RAG, strikes a balance between accuracy and diversity, thus neutralizing the more expressive language generated by the untuned general models. Overall, SoccerComment performs better than specialized soccer commentary generation models in terms of accuracy, semantic consistency, expression diversity, and sentiment expression. In contrast, untuned general video understanding models cannot produce structured and accurate commentary.

Table 2. Effect of Different Modalities for MLLM Training.

Modalities	BLEU	METEOR	Rouge	CIDEr
V	30.12	17.01	29.58	34.14
V+A	30.33	17.64	29.11	35.40
V+A+M	**33.42**	**18.98**	**31.78**	**36.58**

V, A, M represent video, audio, meta-data modality respectly.

All results were obtained with complete RAG-finetuning and inference.

Table 3. Effect of RAG strategy.

Modalities	BLEU	METEOR	Rouge	CIDEr
V	31.88	16.01	30.29	33.09
V+A	33.03	17.34	30.73	35.08
V+A+M	**33.42**	**18.98**	**31.78**	**36.58**
V+A+M ⋆	20.34	15.11	17.34	13.88

⋆ represents the effect of without RAG finetuning.

All results were obtained with complete multi-modal training and inference.

4.3. Ablation Study

4.3.1 Effect of Different Modalities for MLLM Training

We compared the effect of whether or not using multi-modal features during training, and evaluated the performance with only visual features (video). From Table 2, experimental results show that relying solely on the visual modality during training does not yield the best performance. Adding the audio modality results in a slight improvement, but the most significant enhancement comes after incorporating metadata. Metadata includes basic information about the match, as well as the transcribed text of the audio in the current match video segment. This differs from annotation, as metadata tends to represent stylistic examples and background knowledge, which enhances the model's understanding of context and emotions. Previous work has highlighted the success of video-text pre-training, and our experiments also confirmed that incorporating text significantly improves model performance.

4.3.2 Effect of RAG strategy

We conducted experiments to compare the effects of RAG with single visual modality and multi modality, as well as the impact of RAG at different stages (Table 3). RAG using only the visual modality performs worse than the multi-modality approach. Additionally, the results of RAG that combine both audio and metadata are slightly better than those that only mix the audio modality. We attribute this to the joint embedding formed by multi-modal data, which allow the retrieval process to simultaneously consider the relationship between text, audio and video, and capture the correlation between them. We also compared the results of adding multi-modal RAG after MLLM training and fine-tuning , versus adding RAG only during the inference stage without finetuning. The results show that the RAG finetune strategy leads to more accurate commentary generation.

We also studied the effect of varying the number of In-Context Learning (ICL) examples during querying. Both training and inference phases used two retrieved instances per generation, with each additional ICL instance contributing around 1,024 tokens for the video sequence and the rest for language dialogue. Although techniques like Rotary Position Embedding (RoPE) and sliding window methods help the model manage longer contexts, the context window (which limits the attention mechanism's range) remains a bottleneck. This limitation leads to diminishing returns in performance as the number of ICL examples increases from one to three. As LLM architectures evolve, we expect larger context windows to allow more effective usages of ICL examples, potentially improving model performance in long-context tasks.

5. Conclusion

In this paper, we introduce SoccerComment, a novel paradigm for soccer commentary generation. We obtain information-rich input for the model by leveraging multi-modal interaction and language generation capabilities of MLLMs, prior match knowledge, and background commentary text. This meta-data, combined with visual and audio modalities from soccer video clips, enables unsupervised clustering and multi-modal RA-ICL across three modalities. Our approach demonstrates excellent zero-shot capabilities, generating accurate and diverse soccer commentary.

Despite its overall strong performance, our model still has room for improvement in METEOR, CIDEr, and sentiment scores, which signals the directions for exploring the multimodal applications in soccer video commentary at the next stage. Since contents supported by multimodal RAG are quite extensive, future research could involve areas such as commentary stylization. Additionally, it is the ambition of SoccerComment to fulfill the gaps in fully automating soccer commentary generation, marching beyond pre-edited short segment commentary and aiming to generate commentary for unedited soccer match videos and incorporate player/team tracking and classification.

References

[1] Satanjeev Banerjee and Alon Lavie. Meteor: An automatic metric for mt evaluation with improved correlation with human judgments. In *Proceedings of the acl workshop on intrinsic and extrinsic evaluation measures for machine translation and/or summarization*, pages 65–72, 2005. 6

[2] Brian Chen, Andrew Rouditchenko, Kevin Duarte, Hilde Kuehne, Samuel Thomas, Angie Boggust, Rameswar Panda, Brian Kingsbury, Rogerio Feris, David Harwath, et al. Multi-modal clustering networks for self-supervised learning from

unlabeled videos. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 8012–8021, 2021. 4

[3] Wenhu Chen, Hexiang Hu, Xi Chen, Pat Verga, and William W Cohen. Murag: Multimodal retrieval-augmented generator for open question answering over images and text. *arXiv preprint arXiv:2210.02928*, 2022. 3

[4] Wenhu Chen, Hexiang Hu, Chitwan Saharia, and William W Cohen. Re-imagen: Retrieval-augmented text-to-image generator. *arXiv preprint arXiv:2209.14491*, 2022. 3

[5] Zhe Chen, Jiannan Wu, Wenhai Wang, Weijie Su, Guo Chen, Sen Xing, Muyan Zhong, Qinglong Zhang, Xizhou Zhu, Lewei Lu, et al. Internvl: Scaling up vision foundation models and aligning for generic visual-linguistic tasks. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 24185–24198, 2024. 6

[6] Anthony Cioppa, Silvio Giancola, Vladimir Somers, Floriane Magera, Xin Zhou, Hassan Mkhallati, Adrien Deliège, Jan Held, Carlos Hinojosa, Amir M Mansourian, et al. Soccernet 2023 challenges results. *Sports Engineering*, 27(2):24, 2024. 1

[7] Alec Cook and Oktay Karakuş. Llm-commentator: Novel fine-tuning strategies of large language models for automatic commentary generation using football event data. *Knowledge-Based Systems*, 300:112219, 2024. 2

[8] Damai Dai, Yutao Sun, Li Dong, Yaru Hao, Shuming Ma, Zhifang Sui, and Furu Wei. Why can gpt learn in-context? language models implicitly perform gradient descent as meta-optimizers. *arXiv preprint arXiv:2212.10559*, 2022. 5

[9] Adrien Deliege, Anthony Cioppa, Silvio Giancola, Meisam J Seikavandi, Jacob V Dueholm, Kamal Nasrollahi, Bernard Ghanem, Thomas B Moeslund, and Marc Van Droogenbroeck. Soccernet-v2: A dataset and benchmarks for holistic understanding of broadcast soccer videos. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 4508–4519, 2021. 2, 5, 6

[10] Qingxiu Dong, Lei Li, Damai Dai, Ce Zheng, Zhiyong Wu, Baobao Chang, Xu Sun, Jingjing Xu, and Zhifang Sui. A survey on in-context learning. *arXiv preprint arXiv:2301.00234*, 2022. 3

[11] Alexey Dosovitskiy. An image is worth 16x16 words: Transformers for image recognition at scale. *arXiv preprint arXiv:2010.11929*, 2020. 4

[12] Alex Falcon, Giuseppe Serra, and Oswald Lanz. A feature-space multimodal data augmentation technique for text-video retrieval. In *Proceedings of the 30th ACM International Conference on Multimedia*, pages 4385–4394, 2022. 3

[13] Yunfan Gao, Yun Xiong, Xinyu Gao, Kangxiang Jia, Jinliu Pan, Yuxi Bi, Yi Dai, Jiawei Sun, and Haofen Wang. Retrieval-augmented generation for large language models: A survey. *arXiv preprint arXiv:2312.10997*, 2023. 3

[14] Sushant Gautam, Cise Midoglu, Saeed Shafiee Sabet, Dinesh Baniya Kshatri, and Pål Halvorsen. Assisting soccer game summarization via audio intensity analysis of game highlights. In *Proceedings of 12th IOE Graduate Conference*, volume 12, pages 25–32, 2022. 1

[15] Sushant Gautam, Cise Midoglu, Saeed Shafiee Sabet, Dinesh Baniya Kshatri, and Pål Halvorsen. Soccer game summarization using audio commentary, metadata, and captions. In *Proceedings of the 1st Workshop on User-centric Narrative Summarization of Long Videos*, pages 13–22, 2022. 1, 2

[16] Silvio Giancola, Mohieddine Amine, Tarek Dghaily, and Bernard Ghanem. Soccernet: A scalable dataset for action spotting in soccer videos. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 1711–1721, 2018. 2

[17] Silvio Giancola, Anthony Cioppa, Adrien Deliège, Floriane Magera, Vladimir Somers, Le Kang, Xin Zhou, Olivier Barnich, Christophe De Vleeschouwer, Alexandre Alahi, et al. Soccernet 2022 challenges results. In *Proceedings of the 5th International ACM Workshop on Multimedia Content Analysis in Sports*, pages 75–86, 2022. 1

[18] Ahmad Hammoudeh, Bastien Vanderplaetse, and Stéphane Dupont. Soccer captioning: dataset, transformer-based model, and triple-level evaluation. *Procedia Computer Science*, 210:104–111, 2022. 1

[19] Yikun Han, Chunjiang Liu, and Pengfei Wang. A comprehensive survey on vector database: Storage and retrieval technique, challenge. *arXiv preprint arXiv:2310.11703*, 2023. 3

[20] Ziniu Hu, Ahmet Iscen, Chen Sun, Zirui Wang, Kai-Wei Chang, Yizhou Sun, Cordelia Schmid, David A Ross, and Alireza Fathi. Reveal: Retrieval-augmented visual-language pre-training with multi-source multimodal knowledge memory. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pages 23369–23379, 2023. 3

[21] Andreas Husa, Cise Midoglu, Malek Hammou, Steven A Hicks, Dag Johansen, Tomas Kupka, Michael A Riegler, and Pål Halvorsen. Automatic thumbnail selection for soccer videos using machine learning. In *Proceedings of the 13th ACM Multimedia Systems Conference*, pages 73–85, 2022. 1

[22] Gabriel Ilharco, Yuan Zhang, and Jason Baldridge. Large-scale representation learning from visually grounded untranscribed speech. *arXiv preprint arXiv:1909.08782*, 2019. 4

[23] Vanita Jain, Fadi Al-Turjman, Gopal Chaudhary, Devang Nayar, Varun Gupta, and Aayush Kumar. Video captioning: a review of theory, techniques and practices. *Multimedia Tools & Applications*, 81(25), 2022. 1

[24] Bin Lin, Bin Zhu, Yang Ye, Munan Ning, Peng Jin, and Li Yuan. Video-llava: Learning united visual representation by alignment before projection. *arXiv preprint arXiv:2311.10122*, 2023. 6

[25] Chin-Yew Lin. Rouge: A package for automatic evaluation of summaries. In *Text summarization branches out*, pages 74–81, 2004. 6

[26] Muhammad Maaz, Hanoona Rasheed, Salman Khan, and Fahad Shahbaz Khan. Video-chatgpt: Towards detailed video understanding via large vision and language models. *arXiv preprint arXiv:2306.05424*, 2023. 6

[27] Cise Midoglu, Steven A Hicks, Vajira Thambawita, Tomas Kupka, and Pål Halvorsen. Mmsys' 22 grand challenge

on ai-based video production for soccer. *arXiv preprint arXiv:2202.01031*, 2022. 1

[28] Cise Midoglu, Saeed Shafiee Sabet, Mehdi Houshmand Sarkhoosh, Mohammad Majidi, Sushant Gautam, Håkon Maric Solberg, Tomas Kupka, and Pål Halvorsen. Ai-based sports highlight generation for social media. In *Proceedings of the 3rd Mile-High Video Conference*, pages 7–13, 2024. 2

[29] Hassan Mkhallati, Anthony Cioppa, Silvio Giancola, Bernard Ghanem, and Marc Van Droogenbroeck. Soccernet-caption: Dense video captioning for soccer broadcasts commentaries. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5074–5085, 2023. 2, 6

[30] OpenAI. Chatgpt-4. `https://www.openai.com/chatgpt`, 2024. Accessed: 2024-07-17. 5

[31] Kishore Papineni, Salim Roukos, Todd Ward, and Wei-Jing Zhu. Bleu: a method for automatic evaluation of machine translation. In *Proceedings of the 40th annual meeting of the Association for Computational Linguistics*, pages 311–318, 2002. 6

[32] Iqra Qasim, Alexander Horsch, and Dilip K Prasad. Dense video captioning: A survey of techniques, datasets and evaluation protocols. *arXiv preprint arXiv:2311.02538*, 2023. 3

[33] Ji Qi, Jifan Yu, Teng Tu, Kunyu Gao, Yifan Xu, Xinyu Guan, Xiaozhi Wang, Bin Xu, Lei Hou, Juanzi Li, et al. Goal: A challenging knowledge-grounded video captioning benchmark for real-time soccer commentary generation. In *Proceedings of the 32nd ACM International Conference on Information and Knowledge Management*, pages 5391–5395, 2023. 1, 2, 6

[34] Alec Radford, Jong Wook Kim, Tao Xu, Greg Brockman, Christine McLeavey, and Ilya Sutskever. Robust speech recognition via large-scale weak supervision. In *International conference on machine learning*, pages 28492–28518. PMLR, 2023. 5

[35] Jiayuan Rao, Haoning Wu, Chang Liu, Yanfeng Wang, and Weidi Xie. Matchtime: Towards automatic soccer game commentary generation. *arXiv preprint arXiv:2406.18530*, 2024. 1, 2, 5, 6

[36] Mehdi Houshmand Sarkhoosh, Sushant Gautam, Cise Midoglu, Saeed Shafiee Sabet, and Pål Halvorsen. Multimodal ai-based summarization and storytelling for soccer on social media. In *Proceedings of the 15th ACM Multimedia Systems Conference*, pages 485–491, 2024. 1

[37] Hugo Touvron, Louis Martin, Kevin Stone, Peter Albert, Amjad Almahairi, Yasmine Babaei, Nikolay Bashlykov, Soumya Batra, Prajjwal Bhargava, Shruti Bhosale, et al. Llama 2: Open foundation and fine-tuned chat models. *arXiv preprint arXiv:2307.09288*, 2023. 4

[38] Ramakrishna Vedantam, C Lawrence Zitnick, and Devi Parikh. Cider: Consensus-based image description evaluation. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 4566–4575, 2015. 6

[39] Hang Zhang, Xin Li, and Lidong Bing. Video-llama: An instruction-tuned audio-visual language model for video understanding. *arXiv preprint arXiv:2306.02858*, 2023. 6

[40] Lianmin Zheng, Wei-Lin Chiang, Ying Sheng, Siyuan Zhuang, Zhanghao Wu, Yonghao Zhuang, Zi Lin, Zhuohan Li, Dacheng Li, Eric Xing, et al. Judging llm-as-a-judge with mt-bench and chatbot arena. *Advances in Neural Information Processing Systems*, 36, 2024. 4

[41] Bin Zhu, Bin Lin, Munan Ning, Yang Yan, Jiaxi Cui, HongFa Wang, Yatian Pang, Wenhao Jiang, Junwu Zhang, Zongwei Li, et al. Languagebind: Extending video-language pretraining to n-modality by language-based semantic alignment. *arXiv preprint arXiv:2310.01852*, 2023. 4